ENTERPRISE SYSTEM AUDIT

Master Technical Specification

The complete reference: quotation engine, business workflow, backend audit, and performance/code-quality detail.

Company Safari Zangu Ventures
Engagement Full-Codebase Enterprise Audit
Date 20 September 2026
Document 6 of 6
Classification Internal — Management & Engineering

Methodology: every finding in this report is drawn directly from the live codebase, database schema, and configuration — file paths and line numbers are cited throughout. No functionality is assumed to exist without direct verification.

1. Business Workflow Audit

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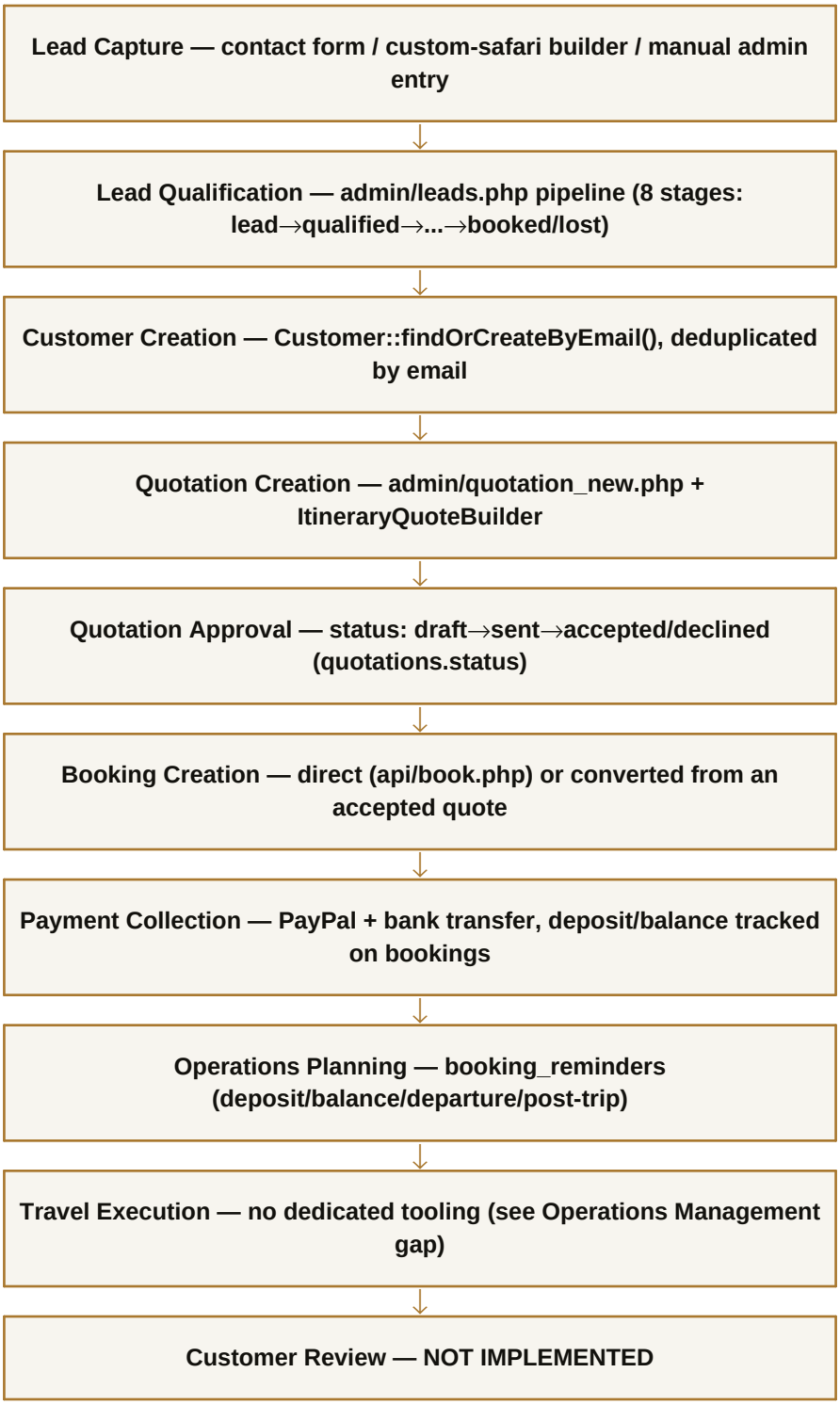

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Repeat Sales — repeat-customer count tracked on admin/customers.php; no formal loyalty automation

1.1 Gap Analysis

Stage	Status	Detail
Lead Capture	✓ SUPPORTED	3 real inbound channels, all wired to the leads/custom_safari_requests tables
Lead Qualification	✓ SUPPORTED	8-stage pipeline with agent assignment and scoring
Customer Creation	✓ SUPPORTED	Automatic dedup by email across all entry points
Quotation Creation/Approval	✓ SUPPORTED	Full itemized builder with PDF/email dispatch
Booking + Payment	✓ SUPPORTED	PayPal + bank transfer; deposit/balance tracked
Operations Planning	■ PARTIAL	Automated reminders exist; no departure manifest, guide/vehicle rostering, or voucher generation
Travel Execution	✗ MISSING	No in-trip tooling of any kind
Customer Review	✗ MISSING	No review/testimonial capture anywhere in the codebase
Repeat Sales	■ PARTIAL	Repeat-customer count is visible; no automated win-back/loyalty flow

2. Quotation Engine — Full Capability Matrix

QUOTATION ENGINE SCORE **3.5 / 10**

Two parallel quotation engines exist — the more capable one is dead code

HIGH

app/Services/QuotationEngine.php implements season multipliers, distinct guide-cost lines, and config-driven airport-transfer/internal-flight line items — genuinely more complete than what's live. Its only HTTP entry point, api/quote.php, has zero callers anywhere in the codebase or frontend JS (grep confirmed). Every real quotation, package price, and booking total is instead computed by app/Services/ItineraryQuoteBuilder.php, a separate, simpler implementation. This is architecturally risky: a future developer fixing a pricing bug in QuotationEngine.php would reasonably assume they'd fixed it everywhere — they wouldn't have.

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EVIDENCE: api/quote.php; grep -rn "quote.php" across all .php/.js (excl. vendor) — zero matches; ItineraryQuoteBuilder.php confirmed as the actual call target from admin/quotation_new.php, admin/package_edit.php, api/book.php, api/itinerary.php
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Pricing Capability	Status	Live-Path Evidence
Park fees (per-park, per-person)	✓ SUPPORTED	ItineraryQuoteBuilder.php:148-149 costs adult/child fees per park per day-entry
Concession fees (distinct from park fees)	✗ MISSING	No distinct concession-fee concept in the live path
Airport/airstrip fees or transfers	■ PARTIAL	Generic 'Other Costs' free-text field, auto-multiplied by pax only if the admin's typed label contains the word 'transfer'
Ground transfers as distinct line item	■ PARTIAL	Same substring-match mechanism as airport fees — not a dedicated feature
Activities (paid add-ons, itemized)	✓ SUPPORTED	activities table feeds an autocomplete; per-day activities itemized and totaled in ItineraryQuoteBuilder.php:162-180
Accommodation costing	■ PARTIAL	Flat rate*nights*rooms only (ItineraryQuoteBuilder.php:126) — rate_per_person, rate_low, rate_high never used; pax count never factored in
Vehicle costing (rate/fuel/driver/maint/insurance)	■ PARTIAL	daily_rate+fuel_cost_day+driver_cost_day pre-bundled into one flat per-day number before it even reaches the calculator — never itemized. maintenance_cost_day/insurance_cost_day (added this engagement) have zero code references anywhere
Guide costs (distinct line item)	■ PARTIAL	Bundled into a manually-typed 'Professional Driver Guide & Service Fee' per-pax figure — no distinct guide concept, no guides table
Driver allowances (distinct from driver_cost_day)	✗ MISSING	No trace anywhere
Fuel costs (own line item)	✗ MISSING	Pre-summed into the bundled vehicle rate before reaching any itemization

Pricing Capability	Status	Live-Path Evidence
Seasonal pricing (real multiplier)	✗ MISSING (live) / PARTIAL (orphaned only)	Zero season references in ItineraryQuoteBuilder.php or quotation_new.php; quotations.season sits at its unused DB default 'peak' forever
Dynamic/demand-based pricing	✗ MISSING	No occupancy/lead-time/demand logic found anywhere
Profit margin / markup	✗ MISSING	packages.price_1pax is computed as raw-cost+VAT with zero markup step. The operational_cost/selling_price/profit_margin_percent columns (added this engagement) are not read or written by any calculation code yet
Agent commissions	✓ SUPPORTED	commission_amount = booking_value * commission_rate/100, triggered on lead pipeline_stage='booked' (api/leads.php:123-124)
Multi-currency	✗ MISSING	PAYPAL_CURRENCY hardcoded 'USD'; GBP_RATE is explicitly display-only per its own code comment, never touches a real total
VAT / tax calculation	■ SUPPORTED, with a code-quality flag	18% applied to post-discount subtotal — but ItineraryQuoteBuilder.php:293 hardcodes the literal 0.18 rather than reading the VAT_RATE config constant (also 0.18 today, but not wired together)
Child pricing	■ PARTIAL	Real distinct child rate ONLY for park fees; accommodation/activities/meals all charge children at the full adult rate; accommodations.child_rate column has zero calculation-code references
Resident (Tanzanian) pricing	✗ MISSING from calculation	park_fees.resident_adult/resident_child exist and are edited via a full admin CRUD form, but ZERO pricing code anywhere selects those columns — every guest is charged the non-resident rate regardless of nationality
Non-resident pricing	✓ SUPPORTED (as the only tier applied)	adult_fee/child_fee are the sole rates used in both engines
EAC citizen pricing	✗ MISSING	No EAC-related column, constant, or code path anywhere in the codebase
Single supplement	✗ MISSING from any total	accommodations.single_supplement column exists, edited via admin CRUD, never read by any pricing calculation

3. Backend Audit

BACKEND SCORE **7.5 / 10**

Genuine service/middleware layer, consistent try/catch discipline; docked for a dead-code parallel pricing engine and 6x-duplicated SMTP setup

3.1 Layer Completeness

Layer	Status	Detail
Routes / Controllers	✓ COMPLETE	34 admin + 30 api entry points, each a real page-controller
Services	✓ COMPLETE	6 real service classes (QuotationEngine*, ItineraryQuoteBuilder, PdfRenderer, Mailer, PayPalClient, ActivityLog*, BookingReminderScheduler)
Models	■ PARTIAL	5 model classes exist but most admin pages bypass them for raw PDO queries — 8 model methods confirmed dead code
Middleware	✓ COMPLETE	Auth, CSRF, RateLimiter, Honeypot — all genuinely applied, not just defined
Validators	■ PARTIAL	Only 1 formal validator class (BookingValidator) — most other endpoints validate inline
Jobs / Queues	■ PARTIAL	email_queue + booking_reminders cover the current need; no general-purpose job system
Policies / Permissions	✓ COMPLETE	Role-based, both nav-level and data-ownership-level scoping verified

** QuotationEngine and ActivityLog are both fully implemented but currently disconnected from real usage — QuotationEngine has no callers; ActivityLog's table has zero rows despite the service existing.*

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4. Performance Audit

PERFORMANCE SCORE **7.2 / 10**

No caching layer anywhere in the application

MEDIUM

No Redis, Memcached, APCu, or file-based cache exists. The only caching-like pattern is a single request-scoped static-variable memoization inside one private method (ItineraryQuoteBuilder.php:318). Every admin dashboard load re-runs all of its queries from zero.

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EVIDENCE: grep for redis|memcache|apcu_|Cache:: across app/admin/api/config/views — zero hits except the one static-var instance
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Recommendation: Not urgent at current traffic/data volume (largest table is 967 rows). Worth a lightweight cache (even a 60-second static-var or file cache) on admin/index.php's dashboard once traffic grows.

N+1 query in the public Packages page

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views/packages.php runs a separate prepared SELECT against the itineraries table inside a foreach loop over packages (once per package with an itinerary_id set), instead of one query with WHERE id IN (...).

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EVIDENCE: views/packages.php:20-26
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Recommendation: Hoist the itinerary fetch above the loop as a single IN() query — trivial fix, only matters once the package catalog grows well beyond 6 rows.

13 admin list pages have no pagination or query limit

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8 admin list pages (bookings, leads, quotations, customers, invoices, custom_requests, activity_log, email_queue) correctly paginate. 13 others run an unbounded SELECT * with no LIMIT at all (accommodations, activities, itineraries, packages, park_fees, seasonal_rates, team_members, users, vehicles, email_templates), or a hardcoded cap with no real pagination (commissions: LIMIT 200, reminders: LIMIT 50/30). Most are currently small reference tables, so this is low-risk today but will degrade silently as data grows, with no page-2 UI to fall back on.

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EVIDENCE: e.g. admin/accommodations.php:50, admin/vehicles.php:47, admin/commissions.php:95, admin/reminders.php:40-53 (full list of 13 files verified)
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Recommendation: Apply the same pagination pattern already used on bookings.php to the remaining list pages before any of these tables grow past a few hundred rows.

Admin dashboard runs 17-30 unbatched, uncached queries per page load LOW

admin/index.php's business-wide dashboard branch runs 17 distinct queries sequentially; the agent-role branch runs 13 more on its own separate code path. None are combined via UNION/subquery where they reasonably could be, and none are cached.

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Recommendation: Low urgency at current scale — every query is a simple COUNT/SUM against small tables. Worth revisiting if the dashboard becomes noticeably slow.

Only 1 of 27 sampled <img> tags sets explicit width/height LOW

19 of 27 images use loading="lazy" correctly, but only 1 sets explicit dimensions — a real layout-shift (CLS) risk on image-heavy pages, though the upload pipeline itself does properly resize/recompress images server-side (max 1920px, quality 82) via a dedicated ImageOptimizer helper, so raw multi-MB phone-camera uploads are not actually a live problem.

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EVIDENCE: app/Helpers/ImageOptimizer.php (comment explicitly documents the original
9MB-upload problem it was built to fix); width/height only on views/about.php:11
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Recommendation: Add explicit width/height (or aspect-ratio CSS) to the remaining image tags — a same-session, low-risk fix.

Background job system is a single DB-table queue, not a real job runner INFO

email_queue (drained every 2 minutes by cron, plus this engagement's new immediate-send-in-request path) and booking_reminders are the only background-processing mechanisms. No Redis-backed queue, no job-class framework, no worker process — adequate for current volume, would need real infrastructure to scale to high email/reminder throughput.

```
EVIDENCE: api/cron/process_email_queue.php; api/cron/process_reminders.php;
app/Mail/Mailer.php
```

Recommendation: Fine as-is at current scale; revisit if outbound email volume grows an order of magnitude.

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5. Code Quality Audit

CODE QUALITY SCORE **7.6 / 10**

SMTP configuration block duplicated 6 times across the codebase

MEDIUM

The isSMTP/Host/Username/Password/SMTPSecure block exists in app/Mail/Mailer.php (the canonical, actually-used version), plus separately in CustomSafariAckMail.php, BookingConfirmationMail.php, api/booking_email.php (still live, linked from admin/bookings.php), and TWICE within api/mailer.php itself.

```
EVIDENCE: app/Mail/Mailer.php:158-166; app/Mail/CustomSafariAckMail.php:15-22;
app/Mail/BookingConfirmationMail.php:21-27; api/booking_email.php:271-279;
api/mailer.php:25-33 and :146-154
```

Recommendation: Refactor all 5 remaining copies to call Mailer::sendRow() (already built and proven this engagement) instead of re-implementing SMTP setup each time.

An entire dead file and two dead classes remain in the codebase

LOW

api/mailer.php is not required/included/linked from anywhere. App\Mail\BookingConfirmationMail and App\Mail\CustomSafariAckMail are never instantiated anywhere — both fully superseded by the Mailer::queue() template system but never removed.

```
EVIDENCE: grep for any reference to api/mailer.php, 'new BookingConfirmationMail', 'new
CustomSafariAckMail' — all zero hits outside their own definitions
```

Recommendation: Safe to delete all three — confirmed genuinely unreferenced, not a risky removal.

8 model methods are dead code

LOW

Formatter::bookingStatusBadge(), TripExtensions::optionsFor(), Booking::all()/count()/totalRevenue()/updatePaymentStatus(), Package::all()/findBySlug(), CustomSafariRequest::all()/count(), TeamMember::all() — none are called anywhere in application code (one is referenced only from a test). Most admin pages query the database directly with raw PDO instead of using these model classes.

```
EVIDENCE: Full list with file:line in the underlying research; each verified via a grep
for external call sites
```

Recommendation: Low priority — either wire the models into the admin pages that currently bypass them (more consistent architecture) or remove the unused methods. Not urgent.

Client-side JS re-implements the server-side pricing calculator for live preview

MEDIUM

admin/quotation_new.php's JS function calcTotals() mirrors ItineraryQuoteBuilder.php's accommodation/vehicle/park/activity/meals/VAT/deposit logic line-for-line, including a duplicated regex heuristic for 'transfer/water/contribution' cost items. The code comments explicitly acknowledge this duplication. It's necessary for an instant client-side preview without a round-trip, but it is two independent implementations of the same math that must be kept in sync by hand.

`EVIDENCE: admin/quotation_new.php (JS calcTotals, ~lines 808-957) vs app/Services/ItineraryQuoteBuilder.php:19-306`

Recommendation: Accept as a deliberate UX tradeoff, but add an automated test that feeds identical inputs to both implementations and asserts matching output, to catch future drift.

Cross-table collation mismatch requires a manual workaround

LOW

bookings uses utf8mb4_unicode_ci while commissions uses utf8mb4_0900_ai_ci — a genuine schema inconsistency that forces admin/reminders.php to explicitly force COLLATE on a join condition to avoid an 'Illegal mix of collations' SQL error.

`EVIDENCE: sql/baseline_schema.sql:158 vs :179; workaround at admin/reminders.php:43,51`

Recommendation: Standardize all tables on one collation (utf8mb4_unicode_ci, since that's what most of the schema already uses) in a dedicated migration, removing the need for ad-hoc COLLATE workarounds.

No TODO/FIXME left in the codebase

GOOD

Exhaustive case-insensitive search across the whole application for TODO/FIXME/XXX markers found zero genuine developer comments — all matches were false positives (placeholder text like 'SZVXXX' in example booking references).

`EVIDENCE: grep -rniE 'TODO|FIXME|XXX' across admin/api/app/config/views/layouts`

Recommendation: No action needed — this is a positive signal about hygiene.

Consistent error handling across sampled admin pages GOOD

8 admin pages sampled in depth (accommodations, bookings, customers, gallery, index, poster_maker, users, vehicles) all wrap every DB call in try/catch, either inline or via the shared sv()/sa() dashboard helpers — no raw unguarded query found that would fatal the page on a DB error.

`EVIDENCE: Per-file try/catch counts verified 1:1 against query counts across the sample`

Recommendation: No action needed.

Consistent snake_case naming throughout the schema GOOD

Every table and column name checked (bookings, commissions, customers, leads, packages) follows snake_case with no camelCase or mixed-case outliers.

`EVIDENCE: sql/baseline_schema.sql, spot-checked 5 tables`

Recommendation: No action needed.

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6. Business Rules Configuration Reference

Every business-rule constant currently governing pricing and operations, for reference:

Constant	Value	Note
DEPOSIT_PERCENT	25%	Deposit required at booking
BALANCE_DAYS	45 days	Balance due before travel
VAT_RATE	18%	Config value — NOT wired to the live calculator, which hardcodes 0.18 separately
SERVICE_CHARGE_RATE	5%	Used only by the orphaned QuotationEngine
CONSERVATION_FEE	$20	Used only by the orphaned QuotationEngine
GBP_RATE	0.75	Display-only conversion, added this engagement, never touches a real total

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7. Closing Note

This document set reflects the codebase and database exactly as they stood on 20 September 2026. Every finding traces to a specific file, line, or live query — re-run the same checks after implementing roadmap items to confirm each one closes cleanly.

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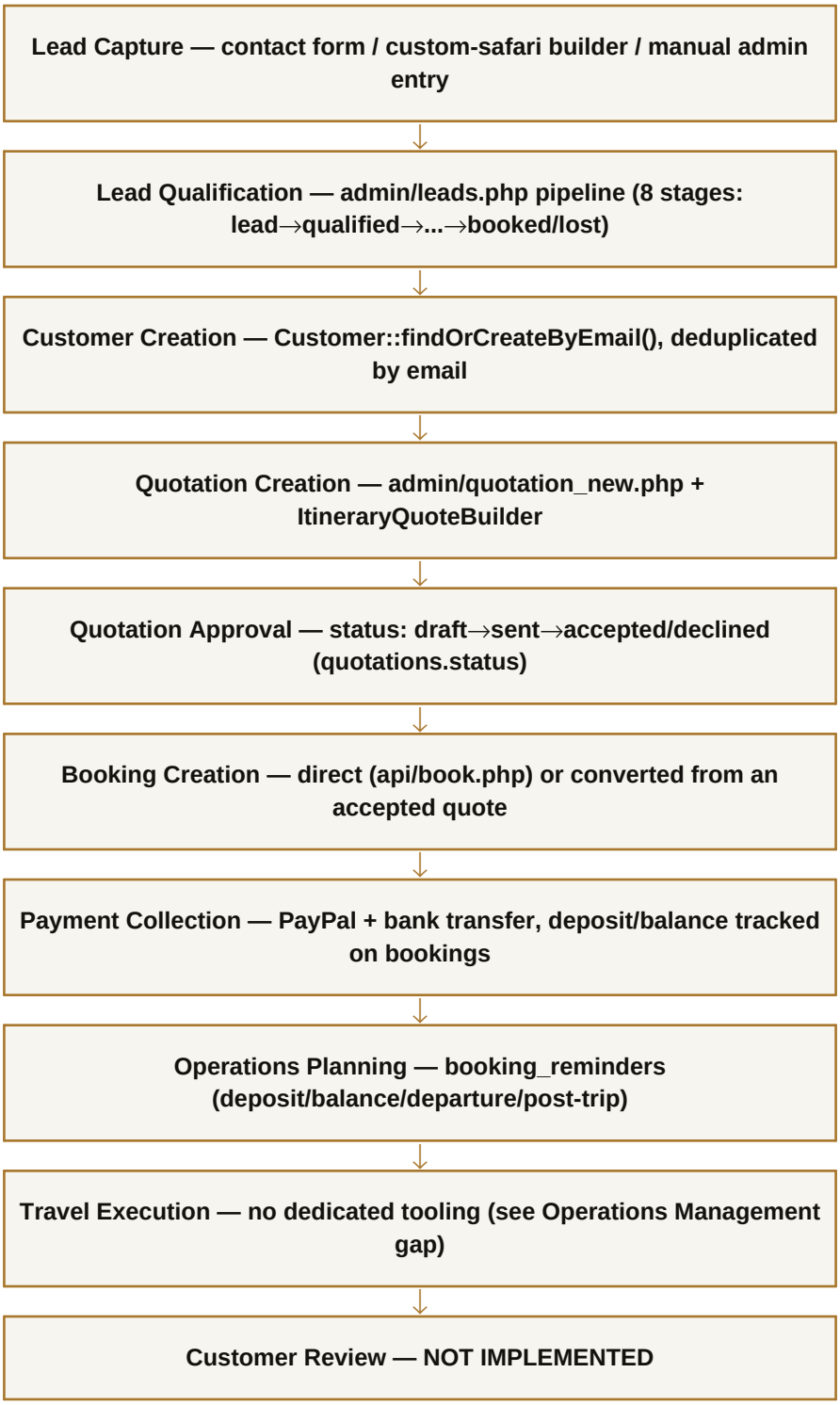

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The isSMTP/Host/Username/Password/SMTPSecure block exists in app/Mail/Mailer.php (the canonical, actually-used version), plus separately in CustomSafariAckMail.php, BookingConfirmationMail.php, api/booking_email.php (still live, linked from admin/bookings.php), and TWICE within api/mailer.php itself.

```
EVIDENCE: app/Mail/Mailer.php:158-166; app/Mail/CustomSafariAckMail.php:15-22;
app/Mail/BookingConfirmationMail.php:21-27; api/booking_email.php:271-279;
api/mailer.php:25-33 and :146-154
```

Recommendation: Refactor all 5 remaining copies to call Mailer::sendRow() (already built and proven this engagement) instead of re-implementing SMTP setup each time.

An entire dead file and two dead classes remain in the codebase

LOW

api/mailer.php is not required/included/linked from anywhere. App\Mail\BookingConfirmationMail and App\Mail\CustomSafariAckMail are never instantiated anywhere — both fully superseded by the Mailer::queue() template system but never removed.

```
EVIDENCE: grep for any reference to api/mailer.php, 'new BookingConfirmationMail', 'new
CustomSafariAckMail' — all zero hits outside their own definitions
```

Recommendation: Safe to delete all three — confirmed genuinely unreferenced, not a risky removal.

8 model methods are dead code

LOW

Formatter::bookingStatusBadge(), TripExtensions::optionsFor(), Booking::all()/count()/totalRevenue()/updatePaymentStatus(), Package::all()/findBySlug(), CustomSafariRequest::all()/count(), TeamMember::all() — none are called anywhere in application code (one is referenced only from a test). Most admin pages query the database directly with raw PDO instead of using these model classes.

```
EVIDENCE: Full list with file:line in the underlying research; each verified via a grep
for external call sites
```

Recommendation: Low priority — either wire the models into the admin pages that currently bypass them (more consistent architecture) or remove the unused methods. Not urgent.

Client-side JS re-implements the server-side pricing calculator for live preview

MEDIUM

admin/quotation_new.php's JS function calcTotals() mirrors ItineraryQuoteBuilder.php's accommodation/vehicle/park/activity/meals/VAT/deposit logic line-for-line, including a duplicated regex heuristic for 'transfer/water/contribution' cost items. The code comments explicitly acknowledge this duplication. It's necessary for an instant client-side preview without a round-trip, but it is two independent implementations of the same math that must be kept in sync by hand.

```
EVIDENCE: admin/quotation_new.php (JS calcTotals, ~lines 808-957) vs
app/Services/ItineraryQuoteBuilder.php:19-306
```

Recommendation: Accept as a deliberate UX tradeoff, but add an automated test that feeds identical inputs to both implementations and asserts matching output, to catch future drift.

Cross-table collation mismatch requires a manual workaround

LOW

bookings uses utf8mb4_unicode_ci while commissions uses utf8mb4_0900_ai_ci — a genuine schema inconsistency that forces admin/reminders.php to explicitly force COLLATE on a join condition to avoid an 'Illegal mix of collations' SQL error.

```
EVIDENCE: sql/baseline_schema.sql:158 vs :179; workaround at admin/reminders.php:43,51
```

Recommendation: Standardize all tables on one collation (utf8mb4_unicode_ci, since that's what most of the schema already uses) in a dedicated migration, removing the need for ad-hoc COLLATE workarounds.

No TODO/FIXME left in the codebase

GOOD

Exhaustive case-insensitive search across the whole application for TODO/FIXME/XXX markers found zero genuine developer comments — all matches were false positives (placeholder text like 'SZVXXX' in example booking references).

```
EVIDENCE: grep -rniE 'TODO|FIXME|XXX' across admin/api/app/config/views/layouts
```

Recommendation: No action needed — this is a positive signal about hygiene.

Consistent error handling across sampled admin pages GOOD

8 admin pages sampled in depth (accommodations, bookings, customers, gallery, index, poster_maker, users, vehicles) all wrap every DB call in try/catch, either inline or via the shared sv()/sa() dashboard helpers — no raw unguarded query found that would fatal the page on a DB error.

```
EVIDENCE: Per-file try/catch counts verified 1:1 against query counts across the sample
```

Recommendation: No action needed.

Consistent snake_case naming throughout the schema GOOD

Every table and column name checked (bookings, commissions, customers, leads, packages) follows snake_case with no camelCase or mixed-case outliers.

```
EVIDENCE: sql/baseline_schema.sql, spot-checked 5 tables
```

Recommendation: No action needed.

6. Business Rules Configuration Reference

Every business-rule constant currently governing pricing and operations, for reference:

Constant	Value	Note
DEPOSIT_PERCENT	25%	Deposit required at booking
BALANCE_DAYS	45 days	Balance due before travel
VAT_RATE	18%	Config value — NOT wired to the live calculator, which hardcodes 0.18 separately
SERVICE_CHARGE_RATE	5%	Used only by the orphaned QuotationEngine
CONSERVATION_FEE	$20	Used only by the orphaned QuotationEngine
GBP_RATE	0.75	Display-only conversion, added this engagement, never touches a real total

7. Closing Note

This document set reflects the codebase and database exactly as they stood on 20 September 2026. Every finding traces to a specific file, line, or live query — re-run the same checks after implementing roadmap items to confirm each one closes cleanly.